Noah Zhang

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LinkedIn Publication

Education

Georgia Institute of Technology Atlanta, GA
Ph.D. in Physics (Advisor: Surabhi Sachdev) *Aug 2023 – Present (Exp. 2028)*
Georgia Institute of Technology Atlanta, GA
M.S. in Computational Science and Engineering (CSE) *Aug 2023 – May 2025*
- **Relevant Coursework:** High Performance Computing, Computational Data Analysis, Modeling & Simulation, Algorithms.

Nanyang Technological University (NTU) Singapore
B.Sc. in Physics (First Class Honors) *Aug 2019 – May 2023*
- **GPA: 4.62/5.00** (Major GPA: 5.00/5.00). Awarded Full Ministry of Education Scholarship.

Technical Skills & Honors

Languages/Libraries: C++ (STL, Boost), Python (NumPy, SciPy, Pandas, **ccxt**, **pykalman**), SQL, LaTeX
Computation: High Performance Computing (HPC), MPI, HTCondor, Git, Linux/Unix
Quantitative: Monte Carlo Simulation (MCMC), Bayesian Inference, Time-Series Analysis, MLE, Signal Processing, Alpha Research (WorldQuant BRAIN)
Honors: PLANCKS Physics Competition (Singapore) — Silver Medalist, 2022 | Odyssey Research Symposium — Best Presentation, 2020

Research & Quantitative Experience

LIGO Scientific Collaboration (Georgia Tech) Aug 2023 – Present
Graduate Research Assistant – Low-Latency Signal Processing *Atlanta, GA*
- **Search Algorithm Optimization:** Co-architected the “Targeted Search” method for GstLAL by integrating Electromagnetic (EM) counterpart data as Bayesian priors. Improved signal detection sensitivity by **30 orders of magnitude** while maintaining the same latency profile.
- **Statistical Validation:** Conducted parameter estimation (PE) review for the SGNL pipeline. Optimized statistical estimators, improving P-P plot (probability-probability) consistency and result reliability by **33%**.
- **Production Analysis:** Contributed to the Intermediate Mass Black Hole (IMBH) offline search for the GWTC-5 production run (data analysis, background statistics).
- **Pipeline Engineering:** Maintain the GstLAL hybrid **C++/Python** pipeline, processing terabytes of streaming data with sub-second latency for real-time trigger generation. Contributed to the Gravitational Wave Transient Catalog (GWTC) study.

Nanyang Technological University May 2020 – Aug 2022
Undergraduate Researcher – Stochastic Modeling & Simulation *Singapore*
- **Algorithmic Speedup:** First author of *Dynamics, Statistics, and Task Allocation of Foraging Ants*. Optimized the agent-based simulation by combining MCMC with a customized Random Walk algorithm.
- **Performance Gain:** Reduced simulation runtime from **days to hours ($\sim$50x speedup)** through code optimization and efficient sampling strategies.

Selected Quantitative Projects

WorldQuant Int’l Quant Championship 2026 – Stage 2, USA Region | *BRAIN, Fast Expression DSL* Mar 2026 – May 2026
- **Overfitting Diagnosis:** Ablated composite alphas to test signal attribution: a fundamental-anchored composite (Sharpe 1.59) collapsed to -0.35 once its price-volume component was removed, showing the alpha had inherited the pool’s PV signal rather than diversifying it. Codified a **per-ingredient standalone-Sharpe check** before any multi-field combination.
- **Systematic Factor Search:** Applied a fixed six-transform template across BRAIN’s data sub-categories, logging **100+ (field $\times$ transform)** simulations to decouple field selection from formula design and turn alpha search into a measurable hit-rate problem.
- **Results:** Submitted **11** cross-sectional alphas on the TOP3000 USA universe (delay-1, subindustry-neutral); the four best-performing held out-of-sample **Sharpe 1.3–2.0** over the full 2019–2023 window (best single alpha **1.98**), turnover 20–45%, max drawdown $<$ 7%, self-correlation $<$ 0.7. Advanced past Stage 1 among **156,000+** registrants.

High-Frequency Market Microstructure Signal Extraction | *Python, Kalman Filter, MLE* Aug 2025 – Feb 2026
- **State-Space Modeling:** Developed a **Kalman Filter** to separate efficient price signals from microstructure noise (bid-ask bounce) in high-frequency Bitcoin tick data, modeling price evolution as a random walk with mean-reverting error.
- **Constrained Optimization:** Implemented **Maximum Likelihood Estimation (MLE)** to calibrate process/measurement noise matrices (Q and R). Diagnosed and resolved regime-switching overfitting by anchoring measurement noise to empirical tick-size volatility.
- **Benchmarking:** Validated custom manual implementation against the **EM algorithm** (via *pykalman*), demonstrating superior signal retention in high-volatility regimes.